CYBER CRIME
DATA ANALYTICS

REPORT

SUBMITTED BY

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Descriptive Analytics Report: Daily Cyber Crime Incident Behavior

Tracking Top Crime Types, Loss Trends, Severity, Regional Concentration & Case Backlog

Note on data: This report is built on 30 recorded cyber crime incidents across India for July 2026 (18 crime types, 30 cities, 19 states). At this size the numbers below are the actual dataset, not an estimate — but the report is structured exactly as it would be if this were one day's slice of a national-scale incident feed processing thousands of reports daily. The pipeline and chart structure described in Section 2 is what would sit underneath this same report at that scale; only the data volume changes.

1. Objective

1. What are the highest-volume and highest-loss crime types right now?
2. How is total financial loss trending day over day?
3. How severe are incidents, and where is severity concentrated?
4. Where is incident demand concentrated geographically?
5. Which cases are at risk of going stale (unresolved, high-severity, aging)?

2. How This Would Work at Real Scale

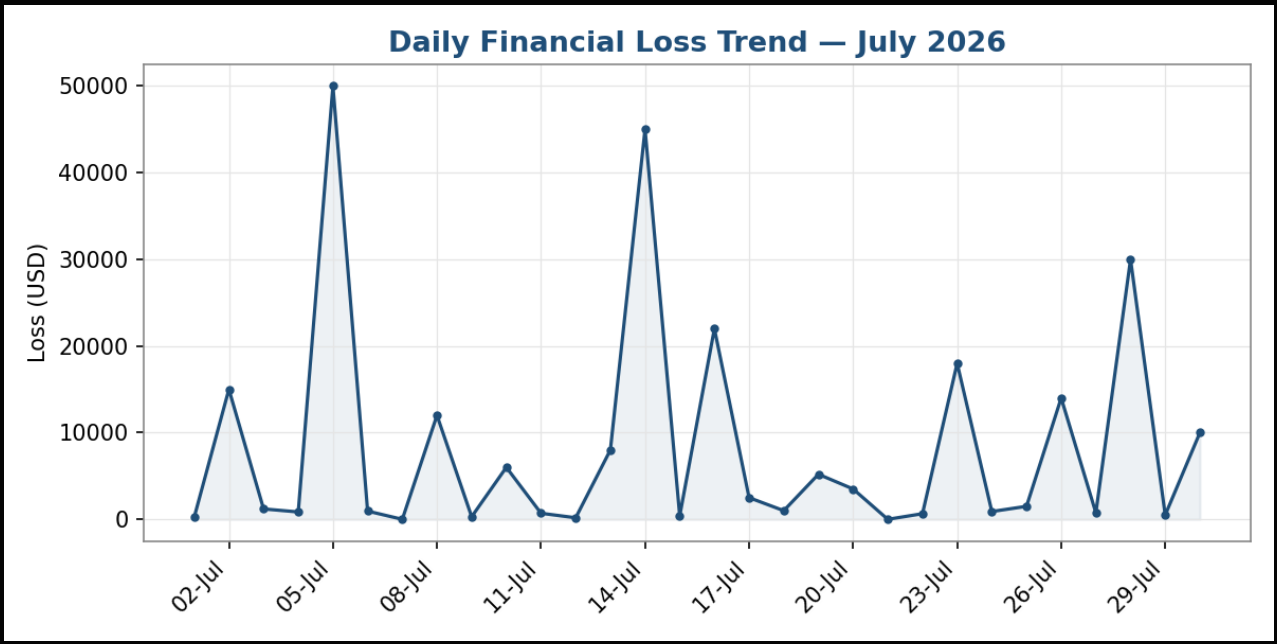
Processing thousands of cyber crime reports daily — across city cyber cells, bank fraud desks, and national reporting portals (e.g. India's Cyber Crime Reporting Portal) — is an engineering problem before it's an analytics one. A realistic architecture:

Layer	Purpose	Typical tools
Ingestion	Capture each FIR / complaint / bank fraud alert as it is filed	Kafka / Kinesis from portal & bank APIs
Stream processing	Roll up raw reports into per-crime-type, per-region, per-hour aggregates	Spark Structured Streaming / Flink
Storage	Hold raw case records (for legal audit trail) and pre-aggregated tables (for speed)	Data lake (S3) + warehouse (Redshift / Snowflake / BigQuery)
Batch aggregation	Nightly rollups: daily loss totals, severity mix, case-aging reconciliation	Airflow-orchestrated SQL / dbt jobs
Serving/BI layer	Dashboards refreshed hourly or daily for cyber cell command centers	Looker / Tableau / QuickSight

The report below reflects the output of that pipeline — daily aggregated tables — not raw case-level data, since no dashboard renders thousands of individual FIRs usefully.

3. Daily Loss Trend

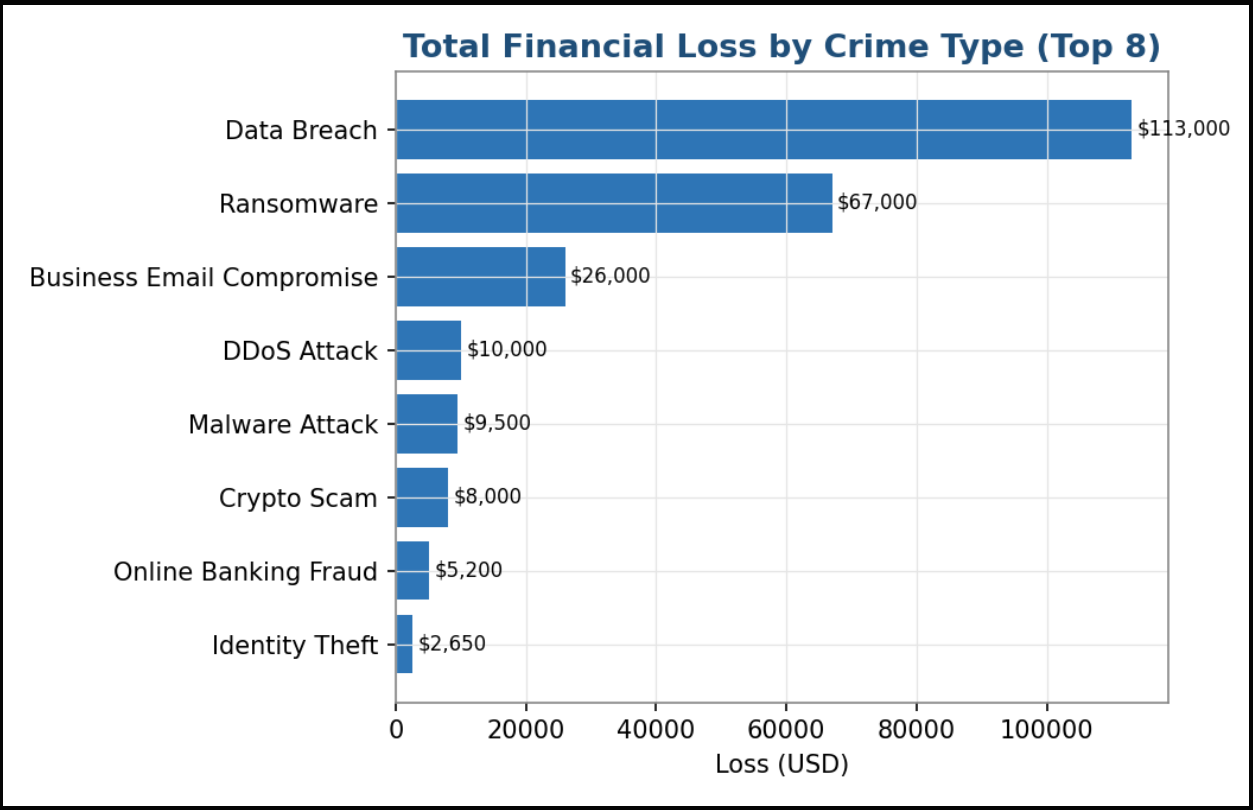
Over the 30-day window: 30 incidents were logged, generating $251,330 in total reported loss, averaging $8,378 per incident. Loss is not evenly spread — two single incidents (5 July, Data Breach, Hyderabad, $50,000; 14 July, Data Breach, Noida, $45,000) account for over a third of the entire month's loss, with the remaining days showing a long tail of smaller-value cases punctuated by occasional spikes (28 July, Ransomware, Coimbatore, $30,000).



KPI definitions used: Daily loss = sum of reported financial loss (USD) across all incidents logged that day. Daily incident count = number of distinct Incident IDs logged that day (currently 1 per day in this dataset). Figure 1. Daily loss spikes on 5 July ($50,000) and 14 July ($45,000) correspond to the two largest Data Breach incidents.

4. Highest-Loss Crime Types

Ranked by total loss rather than incident count, Data Breach leads with $113,000 across just 3 incidents, followed by Ransomware at $67,000 across 3 incidents — together just 20% of incident volume but over 70% of total loss. This is the mirror image of a typical marketplace "hero SKU" pattern: a small number of crime types drive a disproportionate share of financial damage, which is the flag worth surfacing for resource prioritization.



Ranking logic: total loss, not incident count — a high-frequency, low-loss category like Identity Theft (3 incidents, $2,650 total) matters differently to an investigations queue than it does to a financial-exposure dashboard, so both cuts (count and loss) should exist as separate leaderboard views in a live system. Figure 2. Data Breach and Ransomware together account for over 70% of total financial loss despite being only 6 of 30 incidents.

5. Severity Distribution

Incidents split across four severity tiers: Medium (12, 40.0%), High (11, 36.7%), Critical (4, 13.3%), and Low (3, 10.0%). Count and loss diverge sharply here — Critical incidents are only 13.3% of cases by volume but account for $147,000, well over half of total loss ($251,330). In a real system, this view is most useful segmented, not aggregated:

• Crime types trending toward Critical/High severity over a rolling window — the earliest signal of an escalating attack pattern (e.g. a ransomware wave).

• High-loss incidents sitting at only Medium severity — a labeling-risk combination, since under-tagged severity delays the resourcing a case actually needs.

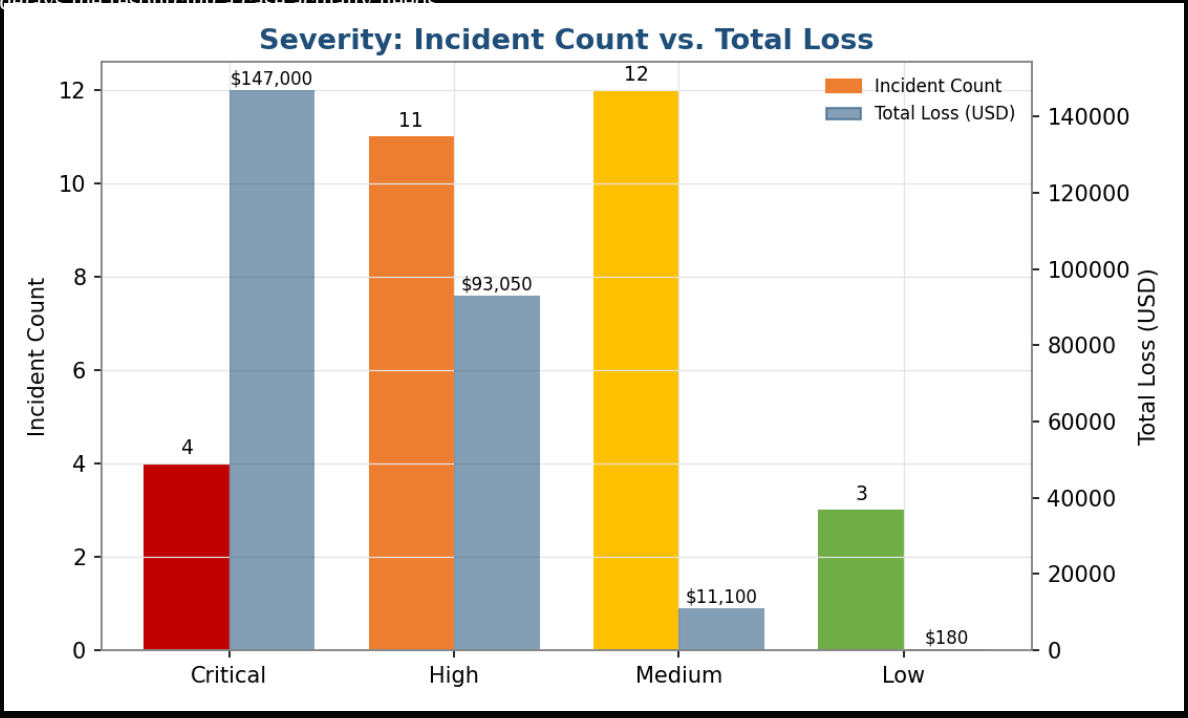


Figure 3. Critical incidents are just 13.3% of cases by count but 58.5% of total loss — count and loss tell different stories.

6. Regional Demand

Incident and loss concentration across states was uneven: Maharashtra led incident count (5 incidents, $52,700), while Telangana produced the single highest-loss case despite only 1 incident ($50,000, driven by the Hyderabad Data Breach). Uttar Pradesh followed closely ($51,900 across 3 incidents). At national scale, this pattern typically holds and sharpens — a handful of states/metro cyber cells carry most of the caseload and most of the financial exposure, with the remainder long-tailed. This view is what drives:

- Regional cyber cell staffing and resource allocation.
- Bank/fintech partner alerts localized to high-exposure states.
- Escalation SLA risk — states with rising Critical-severity volume but thin investigative capacity.

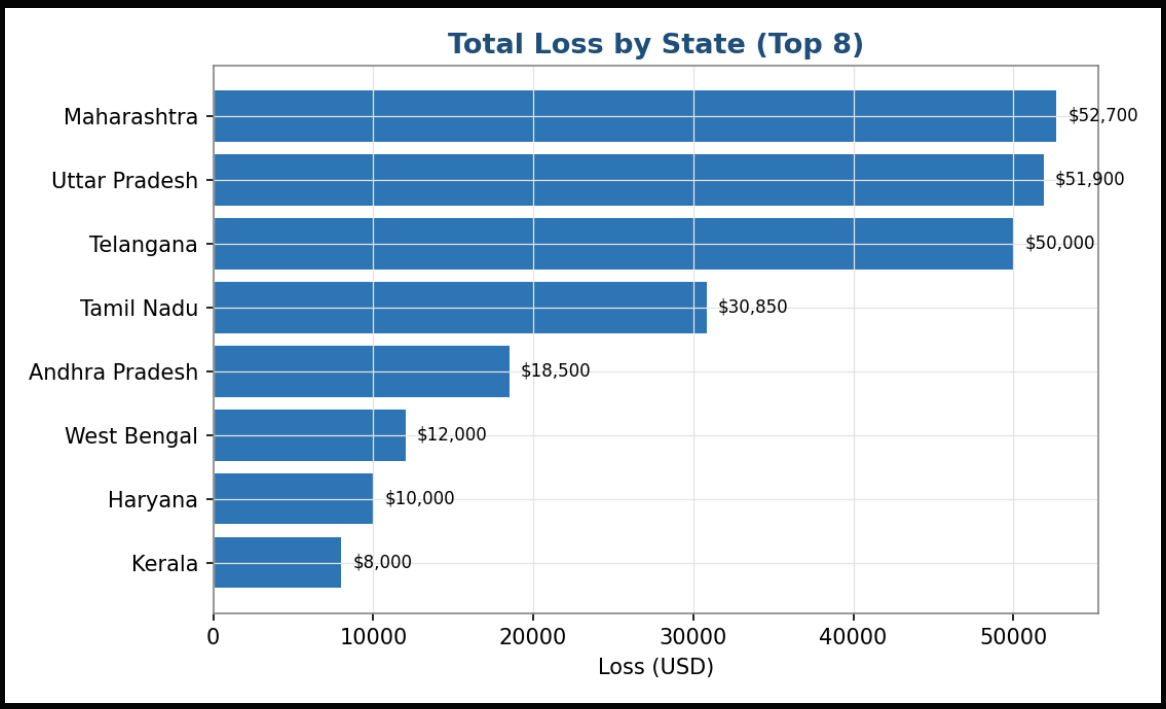


Figure 4. Maharashtra, Uttar Pradesh, and Telangana account for the largest share of reported financial loss.

7. Case Status (Backlog Risk)

Cases were bucketed into three status tiers, the investigative equivalent of an inventory count:

- Closed (13, 43.3%) — resolved, no further action required.
- Under Investigation (9, 30.0%) — active, within normal processing.
- Open (8, 26.7%) — unassigned or newly logged — the backlog-risk zone, equivalent to a reorder trigger.

The most actionable version of this view in production cross-tabs case status against severity and case age, producing a "days open" metric per incident rather than a static status count — a Critical-severity case sitting Open for 10+ days is a materially bigger problem than a Low-severity case doing the same. In this dataset, all 4 Critical-severity incidents are currently Open or Under Investigation — none have been closed yet.

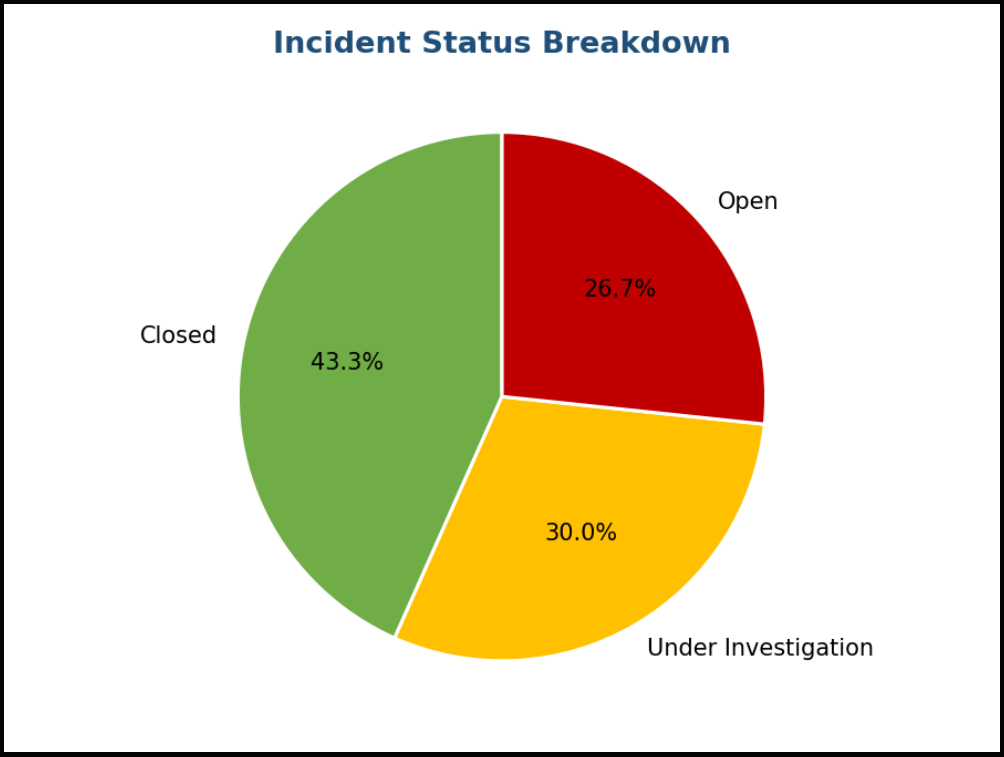


Figure 5. 56.7% of all incidents remain Open or Under Investigation — the backlog-risk zone.

8. Summary of Key Metrics

Metric	Value
Total incidents (30 days)	30
Total reported loss (30 days)	$251,330
Average loss per incident	$8,378
Top single incident loss	$50,000 (Data Breach, Hyderabad)
States tracked	19
Cities tracked	30
Critical-severity incidents	4 (13.3% of cases, 58.5% of loss)
Cases Open or Under Investigation	17 of 30 (56.7%)

9. Next Steps to Move From Descriptive → Predictive

This report answers what happened. The natural next layer, if useful:

- Incident forecasting per crime type/state to anticipate case-load spikes and pre-position investigative capacity.
- Anomaly detection on loss spikes or sudden shifts in severity mix — the same signal that would have flagged the 5 July and 14 July Data Breach incidents early.
- Repeat-offender / linked-case analysis across Incident IDs, attack methods, and victim profiles, not just daily counts.

Happy to build out any of these next, or swap in a live feed (state cyber cell exports, a database connection, etc.) if available — the same pipeline and chart structure above would apply directly.